Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building backend services, full-stack product workflows, and AI-assisted systems using Python, JavaScript/TypeScript, SQL, and cloud-hosted integration pipelines.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and ETL-style workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, schema updates, and maintainable release pipelines across cloud-hosted systems and structured data operations.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and support for shared data platforms.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and multimodal LLM workflows** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and model-backed evaluation pipelines to improve reliability and tooling for production users.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading features, documenting fixes and reducing feedback turnaround by **60%** while improving issue isolation, iteration speed, and maintainability across product workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, ETL workflows, and reliable release coordination.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions while strengthening defect isolation, reproducibility, and documentation across build and integration cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows while strengthening user-facing responsiveness, service coordination, and high-volume application reliability.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and engineering handoffs during debugging and iterative feature delivery.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask APIs and real-time messaging flows** using Firebase Cloud Messaging for 50K+ records, delivering responsive application behavior while supporting webhook integrations, token-based access control, and documented client-facing workflows.
- Integrated **payment and notification features** with maintainable backend logic and technical documentation, reducing manual handling by **70%** while improving operational coordination, usability, and alignment across client and engineering teams.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
Backend & Integrations: REST APIs, GraphQL familiarity, SOAP familiarity, Flask, Node.js, WebSockets, server-side logic
Frontend: React, Next.js, HTML/CSS, user-facing workflows, full-stack product development
Data & Cloud: PostgreSQL, MySQL, MongoDB, SQL/NoSQL, AWS Glue/S3, Azure SQL, ETL workflows, Docker, Jenkins, Git/GitHub, CI/CD, Agile

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **AI-powered document platform** using embeddings, FAISS, Pinecone, and PostgreSQL to support semantic search, summarization, and collaborative workflows across 50K+ records with low-latency retrieval and structured data handling.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management from a unified interface while working across Java backend and TypeScript frontend modules.

HMIS Live | *MERN, Twilio, Sentry, TypeScript*
- Built a **full-stack operational platform** with appointment workflows, consent-based access, notifications, and monitoring, reducing administrative workload by **30%** while improving internal coordination, responsiveness, and day-to-day workflow clarity.